

Q1.

Dimitra is an athlete who competes in 400 m races. The times, in seconds, for her first six races of the 2012 season were

54.86 53.09 53.75 52.88 51.97 51.81

- (a) Assuming that these data form a random sample from a normal distribution, construct a 95% confidence interval for the mean time of Dimitra's races in the 2012 season, giving the limits to two decimal places.

(5)

- (b) For the 2011 season, Dimitra's mean time for her races was 53.41 seconds. After her first six races of the 2012 season, her coach claimed that the data showed that she would be more successful in races during the 2012 season than during the 2011 season. Make **two** comments about the coach's claim.

(2)

(Total 7 marks)

Q2.

A random sample of 50 full-time university employees was selected as part of a higher education salary survey.

The annual salary in thousands of pounds, x , of each employee was recorded, with the following summarised results.

$$\sum x = 2290.0 \quad \text{and} \quad \sum (x - \bar{x})^2 = 28\,225.50$$

Also recorded was the fact that 6 of the 50 salaries exceeded £60 000.

- (a) (i) Calculate values for $\bar{x}$ and s , where s^2 denotes the unbiased estimate of σ^2 .

(3)

- (ii) Hence show why the annual salary, X , of a full-time university employee is unlikely to be normally distributed. Give numerical support for your answer.

(2)

- (b) Assuming that the mean annual salary, $\bar{X}$, of a random sample of 50 full-time university employees is normally distributed, construct a 99% confidence interval for the mean annual salary of full-time university employees.

(4)

- (c) It is claimed that the annual salaries of full-time university employees have an average which exceeds £55 000 and that more than 25% of such salaries exceed £60 000.

Comment on **each** of these two claims.

(3)

(Total 12 marks)

Q3.

Becky owns a taxi. Each weekday morning, she collects Steve from his home and takes

him to the train station.

A record of the times, x minutes, for a random sample of 65 such taxi journeys is summarised by

$$\sum x = 1326.0 \quad \text{and} \quad \sum (x - \bar{x})^2 = 400.24$$

- (a) (i) Calculate the value of the sample mean, $\bar{x}$. (1)
- (ii) Show that, correct to two decimal places, $s = 2.50$, where s^2 denotes the unbiased estimate of the population variance. (2)
- (b) Construct a 96% confidence interval for the mean journey time. (4)
- (c) Comment on Becky's claim that the mean journey time is more than 20 minutes. (2)
- (Total 9 marks)

Q4.

The volume, X litres, of orange juice in a 1-litre carton may be modelled by a normal distribution with unknown mean m .

The volumes, x litres, recorded to the nearest 0.01 litre, in a random sample of 100 cartons are shown in the table.

Volume (x litres)	Number of cartons (f)
0.95 – 0.97	2
0.98 – 1.00	7
1.01 – 1.03	15
1.04 – 1.06	32
1.07 – 1.09	22
1.10 – 1.12	14
1.13 – 1.15	7
1.16 – 1.18	1
Total	100

- (a) For the group '0.98 – 1.00':
- (i) show that it has a mid-point of 0.99 litres;
- (ii) state the minimum and the maximum values of x that could be included in this group. (2)
- (b) Calculate, to three decimal places, estimates of the mean and the standard

deviation of these 100 volumes.

(3)

- (c) (i) Construct an approximate 99% confidence interval for m .

(4)

- (ii) Give a reason why the confidence interval is approximate rather than exact.

(1)

- (d) Give a reason in support of the claim that:

- (i) $\mu > 1$;

- (ii) $P(0.94 < X < 1.16)$ is approximately 1.

(2)

(Total 12 marks)

Q5.

Rice that can be cooked in microwave ovens is sold in packets which the manufacturer claims contain a mean weight of more than 250 grams of rice.

The weight of rice in a packet may be modelled by a normal distribution.

A consumer organisation's researcher weighed the contents, x grams, of each of a random sample of 50 packets. Her summarised results are:

$$\bar{x} = 251.1 \quad \text{and} \quad s = 1.94$$

- (a) (i) Construct a 96% confidence interval for the mean weight of rice in a packet, giving the limits to one decimal place.

(4)

- (ii) Hence comment on the manufacturer's claim.

(2)

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- (b) (i) Construct an interval within which approximately 96% of the weights of rice in individual packets will lie.

(2)

- (ii) The statement '250 grams' is printed on each packet.

Comment on what your interval in part (b)(i) reveals about this statement.

(1)

(Total 9 marks)

Q6.

Rice that can be cooked in microwave ovens is sold in packets which the manufacturer claims contain a mean weight of more than 250 grams of rice.

The weight of rice in a packet may be modelled by a normal distribution.

A consumer organisation's researcher weighed the contents, x grams, of each of a random sample of 50 packets. Her summarised results are:

$$\bar{x} = 251.1 \quad \text{and} \quad \sum (x - \bar{x})^2 = 184.5$$

- (a) Show that, correct to two decimal places, $s = 1.94$, where s^2 denotes the unbiased estimate of the population variance.

(1)

- (b) (i) Construct a 96% confidence interval for the mean weight of rice in a packet, giving the limits to one decimal place.

(4)

- (ii) Hence comment on the manufacturer's claim.

(2)

- (c) The statement '250 grams' is printed on each packet.

Explain, with reference to the values of $\bar{x}$ and s , why the consumer organisation may consider this statement to be dubious.

(2)

(Total 9 marks)

Q7.

The volume, x litres, of magnolia emulsion paint in each of 60 identical containers is measured. The following statistics are calculated from the results.

$$\text{Mean, } \bar{x} = 10.191 \quad \text{Standard deviation, } s = 0.13759$$

- (a) Stating a necessary assumption about the 60 containers, construct a 99 % confidence interval for the mean volume of magnolia emulsion paint in such containers, giving the limits to two decimal places.

(5)

- (b) Labels on the containers indicate a volume of 10 litres of magnolia emulsion paint.

Comment on the likely accuracy of this labelling.

(2)

(Total 7 marks)

Q8.

Vernon, a service engineer, is expected to carry out a boiler service in one hour.

One hour is subtracted from each of his actual times, and the resulting differences, x minutes, for a random sample of 100 boiler services have a mean, $\bar{x}$, of 1.90 and a standard deviation, s , of 3.32 .

- (a) Deduce, in minutes, the mean and the standard deviation of Vernon's actual service times for this sample.

(3)

- (b) Construct a 98% confidence interval for the mean time taken by Vernon to carry out a boiler service.

(4)

- (c) Vernon claims that, on average, a boiler service takes much longer than an hour.

Comment, with a justification, on this claim.

(1)

(Total 8 marks)

Q9.

Vernon, a service engineer, is expected to carry out a boiler service in one hour.

One hour is subtracted from each of his actual times, and the resulting differences, x minutes, for a random sample of 100 boiler services are summarised in the table.

Difference	Frequency
$-6 \leq x < -4$	4
$-4 \leq x < -2$	9
$-2 \leq x < 0$	13
$0 \leq x < 2$	27
$2 \leq x < 4$	21
$4 \leq x < 6$	15
$6 \leq x < 8$	7
$8 \leq x \leq 10$	4
Total	100

- (a) (i) Calculate estimates of the mean and the standard deviation of these differences.

(4)

- (ii) Hence deduce, in minutes, estimates of the mean and the standard deviation of Vernon's actual service times for this sample.

(3)

- (b) (i) Construct an approximate 98% confidence interval for the mean time taken by Vernon to carry out a boiler service.

(4)

- (ii) Give a reason why this confidence interval is approximate rather than exact.

(1)

- (c) Vernon claims that, more often than not, a boiler service takes more than an hour and that, on average, a boiler service takes much longer than an hour.

Comment, with a justification, on **each** of these claims.

(2)

Q10.

A very popular play has been performed at a London theatre on each of 6 evenings per week for about a year. Over the past 13 weeks (78 performances), records have been kept of the proceeds from the sales of programmes at each performance. An analysis of these records has found that the mean was £184 and the standard deviation was £32.

- (a) Assuming that the 78 performances may be considered to be a random sample, construct a 90% confidence interval for the mean proceeds from the sales of programmes at an evening performance of this play.

(4)

- (b) Comment on the likely validity of the assumption in part (a) when constructing a confidence interval for the mean proceeds from the sales of programmes at an evening performance of:

- (i) this particular play;
(ii) any play.

(3)

(Total 7 marks)

Q11.

- (a) A sample of 50 washed baking potatoes was selected at random from a large batch. The weights of the 50 potatoes were found to have a mean of 234 grams and a standard deviation of 25.1 grams.

Construct a 95% confidence interval for the mean weight of potatoes in the batch.

(4)

- (b) The batch of potatoes is purchased by a market stallholder. He sells them to his customers by allowing them to choose any 5 potatoes for £1.

Give a reason why such chosen potatoes are unlikely to represent a random sample from the batch.

(1)

(Total 5 marks)

Q12.

When an alarm is raised at a market town's fire station, the fire engine cannot leave until at least five fire-fighters arrive at the station. The call-out time, X minutes, is the time between an alarm being raised and the fire engine leaving the station.

The value of X was recorded on a random sample of 50 occasions. The results are summarised below, where $\bar{x}$ denotes the sample mean.

$$\sum x = 286.5 \qquad \sum (x - \bar{x})^2 = 45.16$$

- (a) Find values for the mean and standard deviation of this sample of 50 call-out times.

(2)

- (b) Hence construct a 99% confidence interval for the mean call-out time. (4)
- (c) The fire and rescue service claims that the station's mean call-out time is less than 5 minutes, whereas a parish councillor suggests that it is more than $6\frac{1}{2}$ minutes.

Comment on **each** of these claims.

(2)

(Total 8 marks)

Q13.

Currants are sold in 1000-gram packets and in 500-gram packets.

- (a) The weights of 1000-gram packets may be assumed to be normally distributed with a mean of 1012 grams and a standard deviation of 5 grams.

Determine the probability that a randomly selected packet weighs:

- (i) less than 1015 grams; (3)
- (ii) more than 1005 grams; (3)
- (iii) between 1005 grams and 1015 grams. (2)

- (b) The weight, y grams, of each of a random sample of fifty 500-gram packets of currants was recorded with the following results, where $\bar{y}$ denotes the sample mean:

$$n = 50 \quad \sum y = 25\,142.5 \quad \sum (y - \bar{y})^2 = 2519.0361$$

Number of packets weighing less than 500 grams = 6

- (i) Construct a 98% confidence interval for the mean weight of 500-gram packets of currants, giving the limits to two decimal places. (6)
- (ii) On each packet it states 'Contents 500 grams'.

Comment on this statement using **both** the given information **and** your confidence interval.

(3)

(Total 17 marks)

Q14.

Each morning, Mustafa either cycles or walks to school.

- (a) The time, X minutes, taken by Mustafa to cycle to school maybe modelled by a

normal random variable with mean 12 and standard deviation 2.5.

Determine:

(i) $P(X < 15)$;

(2)

(ii) $P(10 < X < 15)$.

(3)

- (b) On each of a random sample of 50 mornings, Mustafa records his time, y minutes, to walk to school. From his recordings, Mustafa calculates the following quantities, where $\bar{y}$ denotes the sample mean.

$$\sum y = 835.0$$

$$\sum (y - \bar{y})^2 = 533.61$$

- (i) Construct a 99% confidence interval for the mean time taken by Mustafa to walk to school.

(6)

- (ii) Mustafa suspects that, on average, the time he takes to walk to school is more than 25% longer than the time he takes to cycle to school.

State, with reasons, whether this suspicion is supported by your confidence interval.

(3)

(Total 14 marks)

Q15.

An internet service provider operates a telephone helpdesk. The random variable T denotes the duration, in minutes, of a telephone call to the helpdesk. The internet service provider recorded the duration, t minutes, of each of a random sample of 50 telephone calls.

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From the recorded durations of these telephone calls, the following values were calculated, where $\bar{t}$ denotes the sample mean.

$$\sum t = 143.50$$

$$\sum (t - \bar{t})^2 = 279.8929$$

- (a) Calculate an unbiased estimate of $\text{Var}(T)$ and hence estimate the standard error of $\bar{T}$.

(3)

- (b) (i) Construct a 99% confidence interval for the mean duration, μ minutes, of telephone calls to the helpdesk.

(5)

- (ii) Hence comment on the claim that $\mu = 3.5$.

(2)

- (c) Three months later, from a random sample of 100 telephone calls, a 99%

confidence interval for μ was determined correctly as (4.12, 5.38).

Indicate, giving reasons, what may now be concluded about the value of μ .

(2)

(Total 12 marks)

Q16.

A machine produces steel rods with lengths that are normally distributed with mean μ and variance σ^2 .

A quality control inspector uses a gauge to measure the length, x centimetres, of each rod in a random sample of 100 rods from the machine's production. The summarised data are as follows.

$$\sum x = 1040.0 \quad \sum x^2 = 11\,102.11$$

- (a) Calculate unbiased estimates of μ and σ^2 .

(3)

- (b) Construct a 99% confidence interval for μ .

(4)

- (c) The gauge used to measure the length is faulty. As a consequence, each measurement taken is 0.2 cm more than the true length.

Use this additional information to write down a revised confidence interval for μ .

(2)

(Total 9 marks)

Q17.

The contents of each of a random sample of 100 cans of a soft drink are measured. The results have a mean of 331.28 ml and a standard deviation of 2.97 ml.

- (a) Show that an unbiased estimate of the population variance is 8.91 ml².

(2)

- (b) Construct a 99% confidence interval for the population mean, giving the limits to two decimal places.

(4)

(Total 6 marks)

Q18.

The volume, X millilitres, of hand cream in tubs is normally distributed with mean μ and variance σ^2 .

- (a) From a random sample of 50 tubs, the following information was determined, where $\bar{x}$ denotes the sample mean.

$$\sum X = 25\,065 \quad \text{and} \quad \sum (X - \bar{x}) = 384.16$$

Construct a 99% confidence interval for μ , giving the limits to two decimal places.

(6)

- (b) It is proposed that, from a second random sample of 50 tubs, **both** a 99% confidence interval and a 90% confidence interval for μ be constructed.

State the probability that **neither** of these confidence intervals will contain μ .

(1)

- (c) It is also proposed that, from a third random sample of 50 tubs, a 99% confidence interval for μ be constructed and that, from a fourth, independent, random sample of 50 tubs, a 90% confidence interval for μ be constructed.

Find the probability that **neither** of these confidence intervals will contain μ .

(2)

(Total 9 marks)

Q19.

- (a) Machine *A* produces plastic balls which have diameters that are normally distributed with a mean of 10.3 cm and a standard deviation of 0.16 cm.

(i) Determine the proportion of balls which have diameters less than 10.5 cm.

(3)

(ii) Calculate the diameter exceeded by 75 per cent of balls.

(4)

- (b) Machine *B* produces beach balls which have diameters that are normally distributed with a mean of μ cm and a standard deviation of 0.24 cm.

The diameter, d cm, of each ball in a random sample of 144 beach balls was measured.

This gave:

$$\Sigma d = 3585.6.$$

(i) Calculate an unbiased estimate of μ .

(1)

(ii) Calculate the standard error of your unbiased estimate.

(2)

(iii) Construct a 95% confidence interval for μ , giving its limits to two decimal places.

(4)

(iv) Hence state, with a reason, whether you agree with a claim that $\mu = 25$.

(2)

(Total 16 marks)

Q20.

An examination of the till roll at a petrol station showed that the previous 400 customers had spent a total of £4256 on petrol and that the standard deviation of the individual amounts spent on petrol was £3.68.

- (a) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer. Assume that the 400 customers may be regarded as a random sample of all customers using this petrol station. (5)
- (b) Further examination of the till roll showed that, although customers could spend any amount they chose on petrol, the great majority of customers spent a multiple of £5. It was suggested that an adequate model for the amount, in £, spent on petrol by customers could be provided by the discrete random variable, X , with the following probability distribution.

x	$P(X = x)$
5	0.15
10	0.63
15	0.15
20	0.07

Calculate:

- (i) the mean of X ;
- (ii) $E(X^2)$;
- (iii) the standard deviation of X .

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- (c) Using the given data and the results of your calculations in parts (a) and (b), comment on the plausibility of X as a model for the amount spent by customers on petrol. (6)
- (d) Miguel, the petrol station owner, decided to offer a £1 token, redeemable on goods at the petrol station, to all customers who spent at least £12 on petrol. He hoped that most of his customers who would have spent £10 on petrol would then spend £15. (3)

The first 200 customers, after the introduction of this offer, spent a total of £2342 on petrol and the standard deviation of the individual amounts spent on petrol was £3.42.

- (i) Using the 5% significance level, examine whether the mean amount spent on petrol by a customer, following the introduction of this offer, exceeds £11. The 200 customers may be regarded as a random sample of all customers using this petrol station. (8)

- (ii) Explain to Miguel the implications of your result in part (d)(i). (2)
- (iii) What further information would you require before advising Miguel whether or not to continue with the offer? (2)
- (Total 26 marks)**

Q21.

An examination of the till roll at a petrol station showed that the previous 400 customers had spent a total of £4256 on petrol and that the standard deviation of the individual amounts spent on petrol was £3.68.

- (a) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer. Assume that the 400 customers may be regarded as a random sample of all customers using this petrol station. (5)
- (b) In the hope of increasing his takings, Miguel, the petrol station owner, decided to offer a £1 token, redeemable on goods at the petrol station, to all customers who spent at least £12 on petrol.

The first 200 customers, after the introduction of this offer, spent a total of £2342 on petrol and the standard deviation of the individual amounts spent on petrol was £3.42.

- (i) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer following the introduction of the offer. Assume that the 200 customers may be regarded as a random sample of all customers using this petrol station. (2)
- (ii) Explain to Miguel the implications of your results in part (a) and part (b)(i). (2)
- (iii) What further information would you require before advising Miguel whether or not to continue with the offer? (2)
- (Total 11 marks)**

Q22.

While examining a water company's finances, an auditor selected a random sample of 90 customers, who owed the company money, in order to scrutinise their accounts. The amounts owed by these 90 customers had a mean of £197 and a standard deviation of £103.

For customers who owed the company money:

- (a) calculate a 95% confidence interval for the mean amount owed; (4)
- (b) state the width of the confidence interval that you have calculated in part (a); (1)

- (c) find the percentage which would be associated with a confidence interval of width £30 calculated from the given data; (5)
- (d) find, approximately, the size of sample which would be required in order to calculate a 99% confidence interval of width £30 for the mean amount owed. (4)

(Total 14 marks)

Q23.

A company manufactures components for the motor industry. The components are designed to have a length of 135.0 mm. Each day a technician takes a random sample of components and calculates a 95% confidence interval for the mean length.

- (a) On a particular Monday the technician takes a sample of nine components. Their lengths, in millimetres, are as follows.

135.1 135.7 134.9 135.2 136.3 135.7 135.9 136.0 135.6

Assuming the lengths of components produced on this Monday may be modelled by a normal distribution, calculate a 95% confidence interval for the mean length. Give your answer to an appropriate degree of accuracy.

(8)

- (b) On a particular Friday a random sample of 60 components had a mean length of 135.8 mm and a standard deviation of 3.9 mm. Calculate a 95% confidence interval for the mean length of components produced on this Friday.

(4)

- (c) The company overhauls all machines when the confidence interval calculated does not contain 135.0 mm. State whether increasing the sample size, n , is likely to increase, decrease or leave unaltered:

- (i) the width of the confidence interval;

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(1)

- (ii) the frequency of machine overhauls, explaining your answer.

(2)

- (d) Using your results in parts (a) and (b) as an example, or otherwise, explain why it is possible that production on a day when the confidence interval does not contain 135.0 mm will sometimes be more satisfactory than production on a day when the confidence interval does contain 135.0 mm.

(2)

(Total 17 marks)

Q24.

A health food co-operative imports a large quantity of dates and packs them into plastic bags labelled 500 grams. George, a Consumer Protection Officer, checked a random sample of the bags and found the weights, in grams, of the contents were as follows.

497 501 486 502 492 508 489 494

- (a) Assuming that weights follow a normal distribution, calculate, for the mean weight of

contents of all the bags:

- (i) a 95% confidence interval; (8)
 - (ii) an 80% confidence interval. (2)
- (b) George is uncertain what conclusion to draw and so decides to start again by taking a much larger sample of the bags. He takes a random sample of 95 bags and finds they have a mean weight of 499.6 grams and a standard deviation of 9.3 grams.
- Use this larger sample to calculate a 90% confidence interval for the mean weight of contents of all the bags. (4)
- (c) The health food co-operative also imports raisins. George intends to take a random sample of 500 gram packets of raisins, weigh the contents and use the results to calculate an 80% and a 95% confidence interval for the mean weight, μ , of the contents of all the co-operative's packets of raisins.
- (i) Find the probability that:
 - (A) the 80% confidence interval contains μ . (1)
 - (B) the 95% confidence interval contains μ but the 80% confidence interval does not. (2)
 - (ii) Instead of calculating both confidence intervals from the same sample, George now decides to calculate the 95% confidence interval from one sample and the 80% confidence interval from a second independent random sample. Find the probability that the 95% confidence interval contains μ but the 80% confidence interval does not. (2)

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(Total 19 marks)

Q25.

Applicants to join a police force are tested for physical fitness. Based on their performance, a physical fitness score is calculated for each applicant. Assume that the distribution of scores is normal.

- (a) The scores for a random sample of ten applicants were
- 55 23 44 69 22 45 54 72 34 66
- Calculate a 99% confidence interval for the mean score of all applicants. (8)
- (b) The scores of a further random sample of 110 applicants had a mean of 49.5 and a standard deviation of 16.5.

Use the data from this second sample to calculate:

- (i) a 95% confidence interval for the mean score of all applicants; (4)
- (ii) an interval within which the score of approximately 95% of applicants will lie. (2)
- (c) By interpreting your results in parts (b)(i) and (b)(ii), comment on the ability of the applicants to achieve a score of 25. (3)
- (d) Give **two** reasons why a confidence interval based on a sample of size 110 would be preferable to one based on a sample of size 10. (2)
- (e) It is suggested that a much better estimate of the mean physical fitness of all recruits could be made by combining the two samples before calculating a confidence interval. Comment on this suggestion. (3)

(Total 22 marks)



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